

## Nmap Scanning Commands with Details

### 1. Basic Host Discovery (Ping Scan)

```
nmap -sn 192.168.1.0/24
```

- -sn: Ping scan (no port scan)
- Scans a subnet to find **live hosts** without checking open ports
- Useful for **host discovery**

### 2. Default Port Scan

```
nmap 192.168.1.1
```

- Scans the **1,000 most common TCP ports**
- Attempts to identify open ports and services

### 3. Full Port Scan (All 65,535 Ports)

```
nmap -p- 192.168.1.1
```

- -p-: Scan **all TCP ports** (1–65535)
- Slower but **thorough**

### 4. Scan Specific Ports

```
nmap -p 22,80,443 192.168.1.1
```

- Scans **only** the ports specified

### 5. Service Version Detection

```
nmap -sV 192.168.1.1
```

- -sV: Attempts to determine **service versions** running on open ports



## 6. OS Detection

nmap -O 192.168.1.1

- -O: Enables **OS fingerprinting**
- May require root privileges



## 7. Aggressive Scan

nmap -A 192.168.1.1

- -A enables:
  - OS detection (-O)
  - Version detection (-sV)
  - Script scanning (-sC)
  - Traceroute
- Very detailed, but **noisy** (easily detected)



## 8. Scan Multiple Hosts/Subnets

nmap 192.168.1.1 192.168.1.2 192.168.2.0/24

- Scan individual IPs or full subnets
- Also accepts input files with -iL:

```
bash
```

```
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```

```
nmap -iL targets.txt
```



## 9. Stealth Scan (SYN Scan)

nmap -sS 192.168.1.1

- -sS: SYN scan (half-open scan)

- Fast and stealthy (may bypass some firewalls/IDS)



## 10. UDP Scan

`nmap -sU -p 53,67,161 192.168.1.1`

- -sU: UDP scan
- Slower than TCP, needed to detect **DNS, DHCP, SNMP**, etc.



## 11. Timing Options (Speed vs Stealth)

`nmap -T4 192.168.1.1`

- -T0 to -T5: Sets timing template
  - T0 = Paranoid (very slow, stealthy)
  - T4 = Fast
  - T5 = Insane (for LAN only)



## 12. Scan Behind Firewalls (Fragmentation)

`nmap -f 192.168.1.1`

- -f: Fragment packets to evade simple firewalls/IDS

**Caution:** May cause issues or alert defenses



## 13. Output Scan Results

`nmap -oN result.txt 192.168.1.1`      # Normal output

`nmap -oX result.xml 192.168.1.1`      # XML format

`nmap -oG result.gnmap 192.168.1.1`      # Grepable output

You can also combine formats:

`nmap -oA scan_results 192.168.1.1`



## 15. Scan IPv6 Targets

```
nmap -6 -sV fe80::1
```

- -6: Enable IPv6 scanning



### Pro Tips

- Use sudo for OS detection and SYN scans.
- Combine -sS -sV -O -T4 -p- for full scans:

```
sudo nmap -sS -sV -O -p- -T4 192.168.1.1
```

- Use --top-ports 100 to scan the top 100 most common ports:

```
nmap --top-ports 100 192.168.1.1
```

The -PR option in Nmap stands for **ARP Ping Scan**. It's a **host discovery method** used to **identify live hosts on a local Ethernet network** by sending **ARP requests**.



### Command Syntax:

```
bash
```

```
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```

```
sudo nmap -PR 192.168.1.0/24
```



### How it Works:

- Nmap sends **ARP requests** to each IP in the target range.
- If a host is up, it replies with an **ARP response** (with its MAC address).

- This method is **very reliable on local networks**, because:
    - ARP is **required** for communication on a LAN
    - It **bypasses firewalls** that block ICMP or TCP pings
-  **Note:** Only works on **local subnets** (within the same Layer 2 broadcast domain). It won't work across routers or on remote networks.